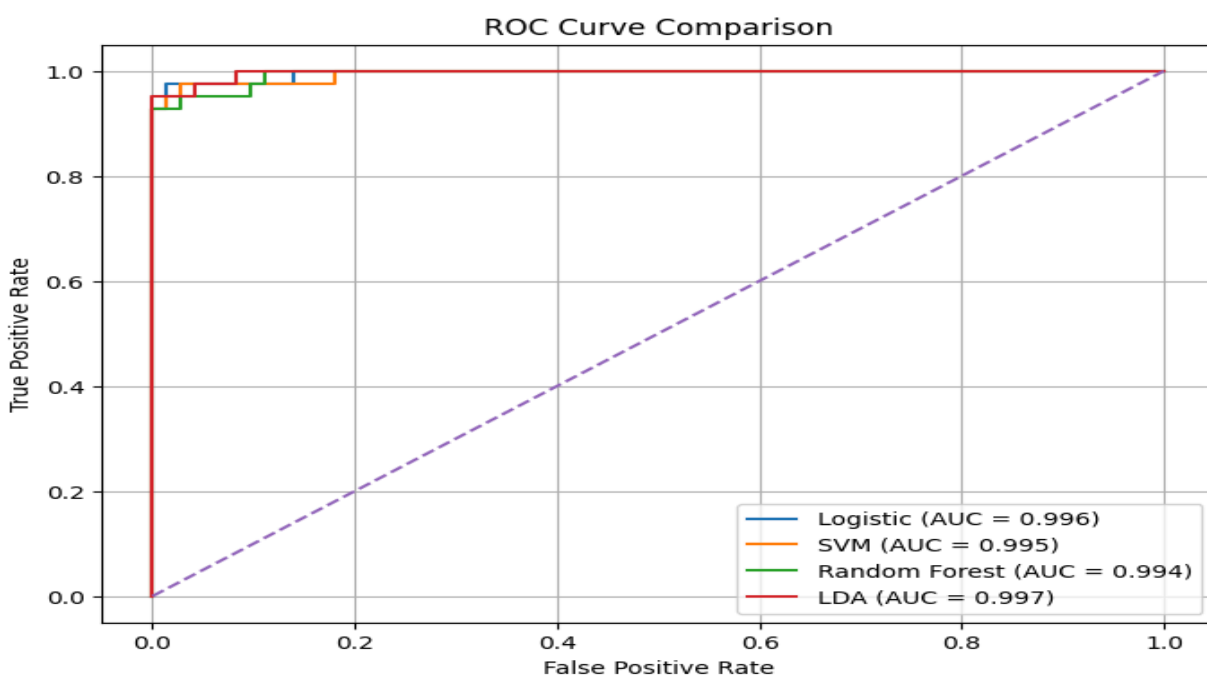


Breast Cancer Prediction using Machine Learning

Results

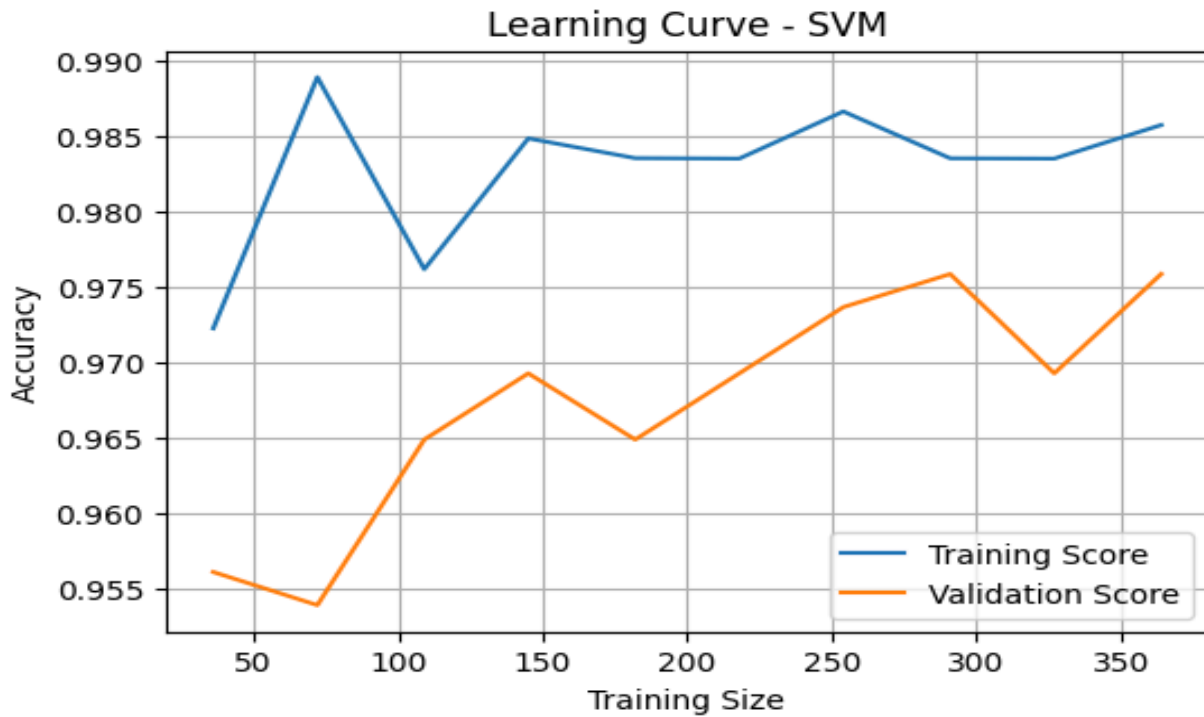
ROC Curve Comparison

All four models score high AUCs, from 0.994 to 0.997. LDA was slightly ahead, followed by Logistic Regression, SVM, and Random Forest. This confirms all four are strong at ranking cases correctly, even though the results table showed LDA trailing on recall at its chosen threshold.



Learning Curve, SVM

Training accuracy stays high and fairly flat around 0.984 to 0.989, while validation accuracy starts lower at 0.956 and climbs steadily as more training data is added, reaching about 0.976 by the end. The gap between the two narrows but doesn't fully close, suggesting the model would likely keep improving slightly with more data, though it's already performing well and shows no signs of serious overfitting.



Learning Curve, Random Forest

Training accuracy sits at a perfect 1.00 across every training size, while validation accuracy starts around 0.94 and climbs gradually to about 0.96. That gap between training and validation is a classic sign the model fits the training data almost perfectly but generalizes a bit less well, some overfitting, though validation performance still holds up reasonably well and continues improving as more data is added.



Feature Importance, Random Forest

Perimeter worst and area worst come out as the two strongest predictors, followed closely by concave points worst and concave points mean, then radius worst. The "worst" measurements, capturing the most extreme values found in each cell nucleus, dominate the top of the list, suggesting the most severe cell abnormalities carry more predictive weight than average measurements across the tumor.

